



Knowledge Graphs

Lecture 5 - Knowledge Graph Applications

5.6 Knowledge Graph Analytics

Prof. Dr. Harald Sack & Dr. Mehwish Alam

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Knowledge Graphs

Lecture 5: Knowledge Graph Applications

5.6 Knowledge Graph Analytics

Knowledge Mining and Knowledge Discovery

Knowledge Discovery

valid novel potentially useful
ultimately understandable patterns

- **valid:** to a certain degree the discovered patterns should also hold for new, previously unseen problem instances.
- **novel:** at least to the system and preferable to the user.
- **potentially useful:** they should lead to some benefit to the user or task.
- **ultimately understandable:** the end user should be able to interpret the patterns either immediately or after some post-processing.

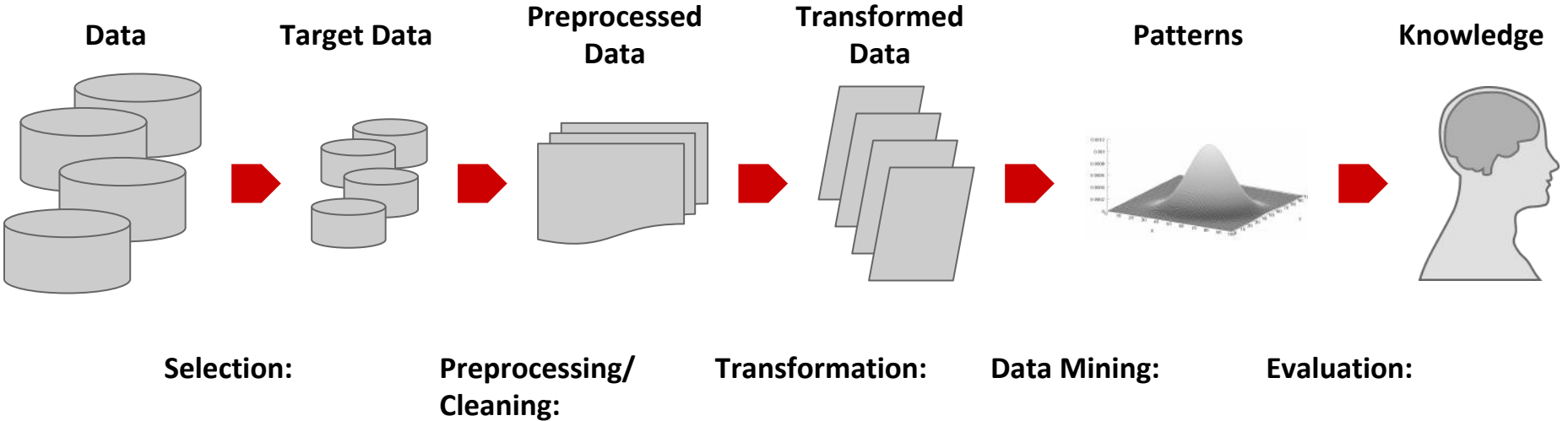
Knowledge Mining and Knowledge Discovery

Knowledge Discovery

valid novel potentially useful
ultimately understandable patterns

- **Goals:**
 - **Descriptive Modelling:** explains the characteristic and the behaviour of the observed data
 - **Predictive Modelling:** predicts the behaviour of new data based on some model
- **Important:**
 - The extracted model/pattern does not have to apply in 100% of the cases

The Knowledge Discovery Process



Hands-On Example - Knowledge Graph Analytics

- -
 - **Data Acquisition**
 - random example
- Physicists
- DBpedia Wikidata



1. Data Acquisition

- random example



WIKIDATA

Statements

instance of	human	edit
	2 references	
+ add value		

image		edit
-------	--	------

1. Data Acquisition



random example

statement

qualifier



country of citizenship	Kingdom of France	edit
	start time21 March 1768 <i>Gregorian</i>	
	end time21 September 1792 <i>Gregorian</i>	
	0 references	+ add reference
	French First Republic	edit
	start time21 September 1792 <i>Gregorian</i>	
	end time18 May 1804 <i>Gregorian</i>	
	0 references	+ add reference
	First French Empire	edit
	start time18 May 1804 <i>Gregorian</i>	
	end time6 April 1814 <i>Gregorian</i>	
	0 references	+ add reference
	Bourbon Restoration	edit
	start time6 April 1814 <i>Gregorian</i>	
	end time20 March 1815 <i>Gregorian</i>	
	0 references	+ add reference
	First French Empire	edit
	start time20 March 1815 <i>Gregorian</i>	
	end time7 July 1815 <i>Gregorian</i>	
	0 references	+ add reference

WikiData Recap

object
=
subject/context
for statement

different namespaces for properties

- **wdt:** _____
wd:Q8772 wdt:P27 ?country .

country of citizenship

Kingdom of France

start time

21 March 1768 *Gregorian*

end time

21 September 1792 *Gregorian*

▼ 0 references

French First Republic

start time

21 September 1792 *Gregorian*

end time

18 May 1804 *Gregorian*

▼ 0 references

First French Empire

start time

18 May 1804 *Gregorian*

end time

6 April 1814 *Gregorian*

▼ 0 references

Bourbon Restoration

start time

6 April 1814 *Gregorian*

end time

20 March 1815 *Gregorian*

▼ 0 references

First French Empire

start time

20 March 1815 *Gregorian*

WikiData Recap

different namespaces for properties

- **wdt:** _____
wd:Q8772 wdt:P27 ?country .
- **p:** _____
wd:Q8772 p:P27 ?country_statement .

country of citizenship

Kingdom of France

start time

21 March 1768 *Gregorian*

end time

21 September 1792 *Gregorian*

▼ 0 references

French First Republic

start time

21 September 1792 *Gregorian*

end time

18 May 1804 *Gregorian*

▼ 0 references

First French Empire

start time

18 May 1804 *Gregorian*

end time

6 April 1814 *Gregorian*

▼ 0 references

Bourbon Restoration

start time

6 April 1814 *Gregorian*

end time

20 March 1815 *Gregorian*

▼ 0 references

First French Empire

start time

20 March 1815 *Gregorian*

statement

WikiData Recap

different namespaces for properties

- **wdt:** _____
wd:Q8772 wdt:P27 ?country .
- **p:** _____
wd:Q8772 p:P27 ?country_statement .
- **pq:** _____
?country_statement pq:P582 ?statement_value

property and
object/value of
statement

country of citizenship	Kingdom of France
start time	21 March 1768 <i>Gregorian</i>
end time	21 September 1792 <i>Gregorian</i>
	▼ 0 references
	French First Republic
start time	21 September 1792 <i>Gregorian</i>
end time	18 May 1804 <i>Gregorian</i>
	▼ 0 references
	First French Empire
start time	18 May 1804 <i>Gregorian</i>
end time	6 April 1814 <i>Gregorian</i>
	▼ 0 references
	Bourbon Restoration
start time	6 April 1814 <i>Gregorian</i>
end time	20 March 1815 <i>Gregorian</i>
	▼ 0 references
	First French Empire
start time	20 March 1815 <i>Gregorian</i>

Knowledge Graph Analytics with SPARQL

-
- **WIKIDATA SPARQL endpoint**
 -
 -
 -
 -
 -

Knowledge Graph Analytics with SPARQL

- what other occupations do Physicists have?


Wikidata Query Service

Examples
Help
More tools

```

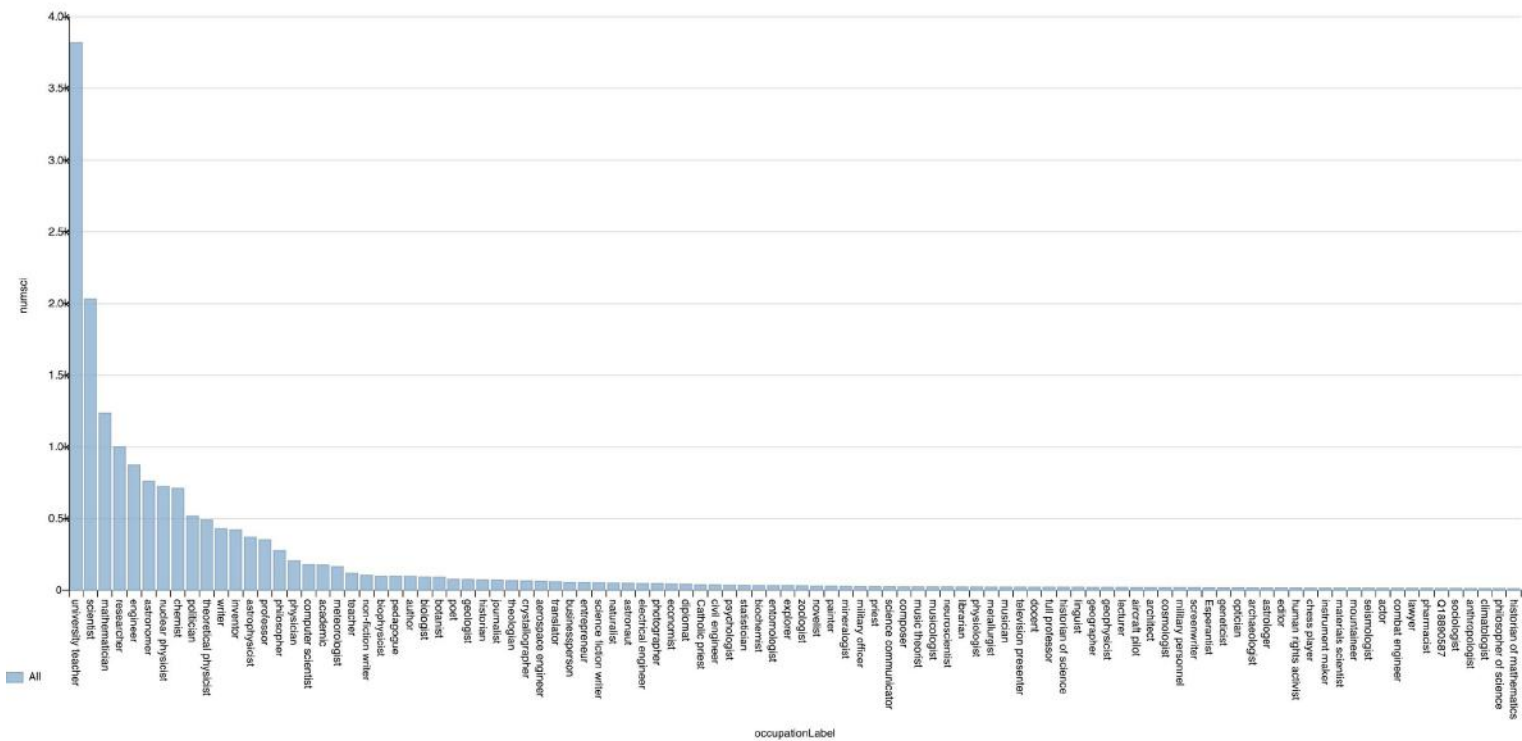
1 #defaultView:BarChart
2 SELECT ?occupationLabel (COUNT(DISTINCT(?scientist)) AS ?numsci) WHERE {
3   ?scientist wdt:P106 wd:Q169470 .
4   ?scientist wdt:P106 ?occupation FILTER (?occupation != wd:Q169470)
5   SERVICE wikibase:label { bd:serviceParam wikibase:language "en, fr, es" }
6 } GROUP BY ?occupationLabel
7 ORDER BY DESC(?numsci)
8 Limit 100

```




Knowledge Graph Analytics with SPARQL

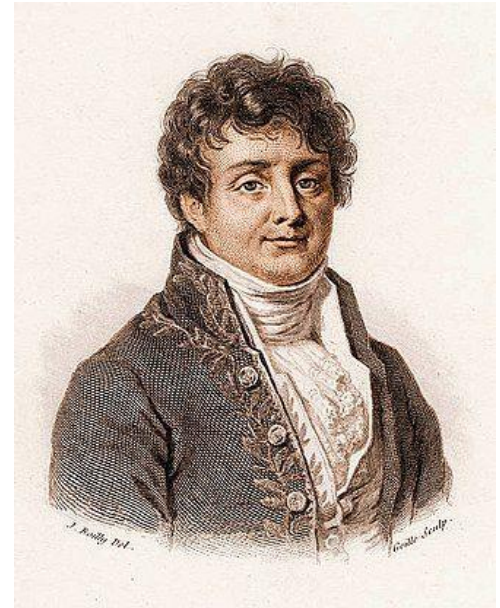
- what other occupations do Physicists have?



2. Get Data

- There is so much more to know about physicists

```
SELECT ?physicist (SAMPLE(YEAR(?birthdate)) AS ?bdate)
  (COUNT(DISTINCT(?country)) AS ?countries)
  (COUNT(DISTINCT(?occupation)) AS ?occupations)
  (COUNT(DISTINCT(?employer)) AS ?employers)
  (COUNT(DISTINCT(?award)) AS ?awards)
  (COUNT(DISTINCT(?member)) AS ?members)
  (COUNT(DISTINCT(?field)) AS ?fields)
  (SAMPLE(?sex) AS ?gender)
WHERE {
  ?physicist wdt:P106 wd:Q169470 .
  ?physicist wdt:P27 ?country .
  ?physicist wdt:P569 ?birthdate .
  ?physicist wdt:P106 ?occupation .
  ?physicist wdt:P108 ?employer .
  ?physicist wdt:P166 ?award .
  ?physicist wdt:P463 ?member .
  ?physicist wdt:P21 ?sex .
  ?physicist wdt:P101 ?field .
} GROUP BY ?physicist
```



2. Get Data

- There is so much more to know about physicists

Wikidata Query Service

```

1 SELECT ?physicist (SAMPLE(YEAR(?birthdate)) AS ?bdate)
2 (COUNT(DISTINCT(?country)) AS ?countries)
3 (COUNT(DISTINCT(?occupation)) AS ?occupations)
4 (COUNT(DISTINCT(?employer)) AS ?employers)
5 (COUNT(DISTINCT(?award)) AS ?awards)
6 (COUNT(DISTINCT(?member)) AS ?members)
7 (COUNT(DISTINCT(?field)) AS ?fields)
8 (SAMPLE(?sex) AS ?gender)
9 WHERE {
10 ?physicist wdt:P106 wd:Q169470 .
11 ?physicist wdt:P27 ?country .
12 ?physicist wdt:P569 ?birthdate .
13 ?physicist wdt:P106 ?occupation .
14 ?physicist wdt:P108 ?employer .
15 ?physicist wdt:P166 ?award .
16 ?physicist wdt:P463 ?member .
17 ?physicist wdt:P21 ?sex .
18 ?physicist wdt:P101 ?field .
19 } GROUP BY ?physicist
20

```

1480 results in 203 ms

Code Download

Search

physicist	bdate	countries	occupations	employers	awards	members	fields	gender
Q937	1879	5	14	15	17	18	1	Q6581097
Q80	1955	1	7	6	40	5	2	Q6581097
Q4517	1911	3	8	1	10	3	2	Q6581097
Q6722	1777	3	8	1	6	12	10	Q6581097
Q675	1775	1	4	2	2	10	1	Q6581097
Q680	1745	4	4	1	3	6	1	Q6581097
Q464	1821	2	4	1	1	3	2	Q6581097
Q1585	1789	2	3	2	3	4	1	Q6581097
Q30693	1822	2	4	4	6	15	1	Q6581097

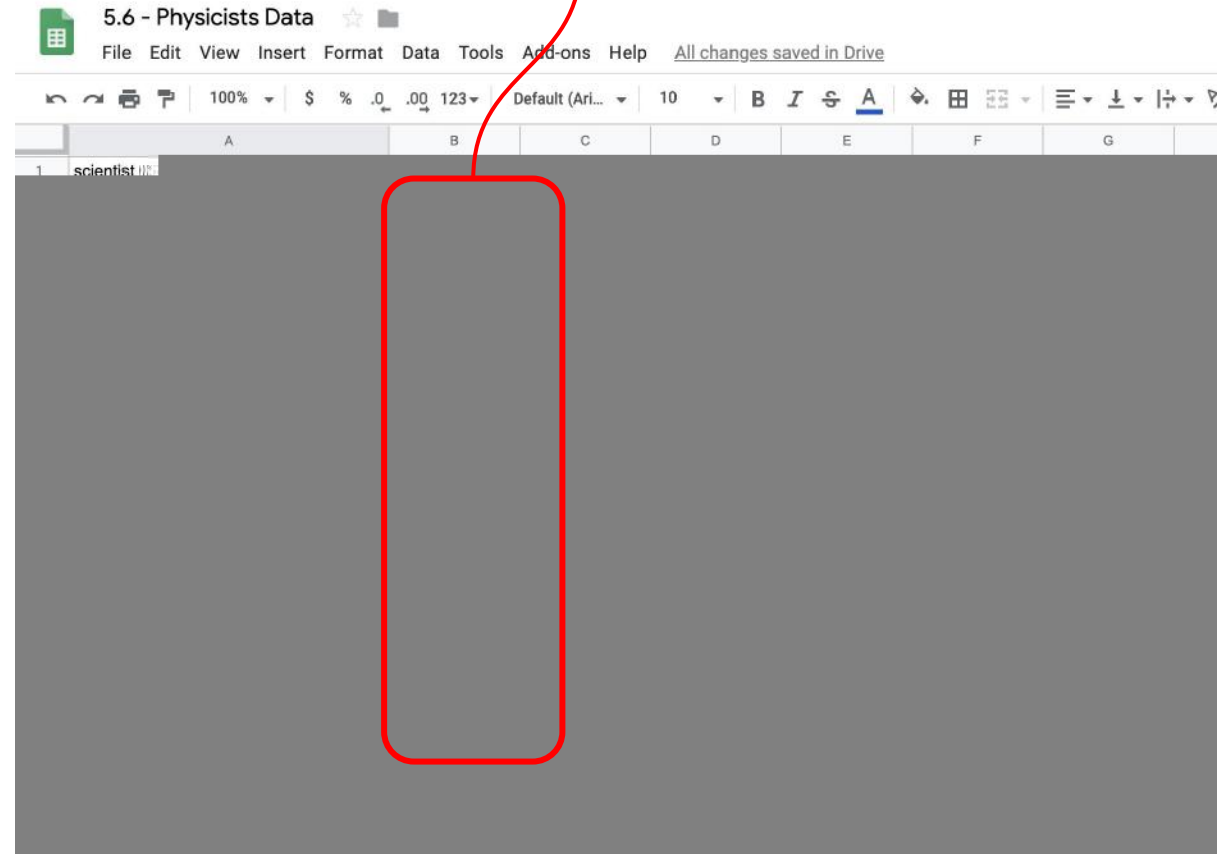
2. Get Data

5.6 - Physicists Data

File Edit View Insert Format Data Tools Add-ons Help All changes saved in Drive								
100% \$ % .0 .00 123 Default (Ari... 10 B I S A								
	A	B	C	D	E	F	G	H
1	scientist	bdate	countries	occupations	employers	awards	members	
2	http://www.wikidata.org/entity/Q80	1955-06-08	1	7	6	40	5	
3	http://www.wikidata.org/entity/Q164384	1881-05-11	6	8	4	37	29	
4	http://www.wikidata.org/entity/Q183279	1930-03-15	2	4	2	33	12	
5	http://www.wikidata.org/entity/Q28189	1926-01-29	2	3	5	32	11	
6	http://www.wikidata.org/entity/Q71013	1941-10-30	2	3	5	31	10	
7	http://www.wikidata.org/entity/Q106624	1946-02-26	2	5	1	30	14	
8	http://www.wikidata.org/entity/Q997	1921-05-21	2	3	1	28	6	
9	http://www.wikidata.org/entity/Q83552	1916-07-11	2	3	6	28	6	
10	http://www.wikidata.org/entity/Q335213	1942-06-23	1	5	4	28	12	
11	http://www.wikidata.org/entity/Q201513	1951-08-26	1	4	1	28	8	
12	http://www.wikidata.org/entity/Q184566	1915-07-28	1	4	3	27	12	
13	http://www.wikidata.org/entity/Q487983	1903-02-13	3	3	3	26	4	
14	http://www.wikidata.org/entity/Q2587259	1946-01-23	2	1	1	26	4	
15	http://www.wikidata.org/entity/Q29573	1930-11-11	1	3	1	25	4	
16	http://www.wikidata.org/entity/Q80917	1962-09-04	1	7	3	25	5	
17	http://www.wikidata.org/entity/Q48983	1901-02-28	1	8	5	23	11	
18	http://www.wikidata.org/entity/Q91410	1962-12-23	2	4	2	23	5	
19	http://www.wikidata.org/entity/Q190697	1933-04-01	1	3	6	23	12	
20	http://www.wikidata.org/entity/Q193803	1931-08-08	1	8	7	23	7	
21	http://www.wikidata.org/entity/Q16389	1933-11-04	2	5	4	22	7	
22	http://www.wikidata.org/entity/Q1030228	1934-03-23	2	4	2	22	10	
23	http://www.wikidata.org/entity/Q148109	1910-10-19	1	5	1	22	8	
24	http://www.wikidata.org/entity/Q1975294	1921-10-15	1	6	2	22	4	
25	http://www.wikidata.org/entity/Q1248892	1934-06-30	2	2	2	22	12	
26	http://www.wikidata.org/entity/Q164396	1908-09-18	3	4	3	22	12	
27	http://www.wikidata.org/entity/Q574605	1925-06-08	2	4	1	21	7	
28	http://www.wikidata.org/entity/Q155794	1906-07-02	2	2	5	21	8	
29	http://www.wikidata.org/entity/Q155794							

3. Clean Up Data

Can we trust this data?

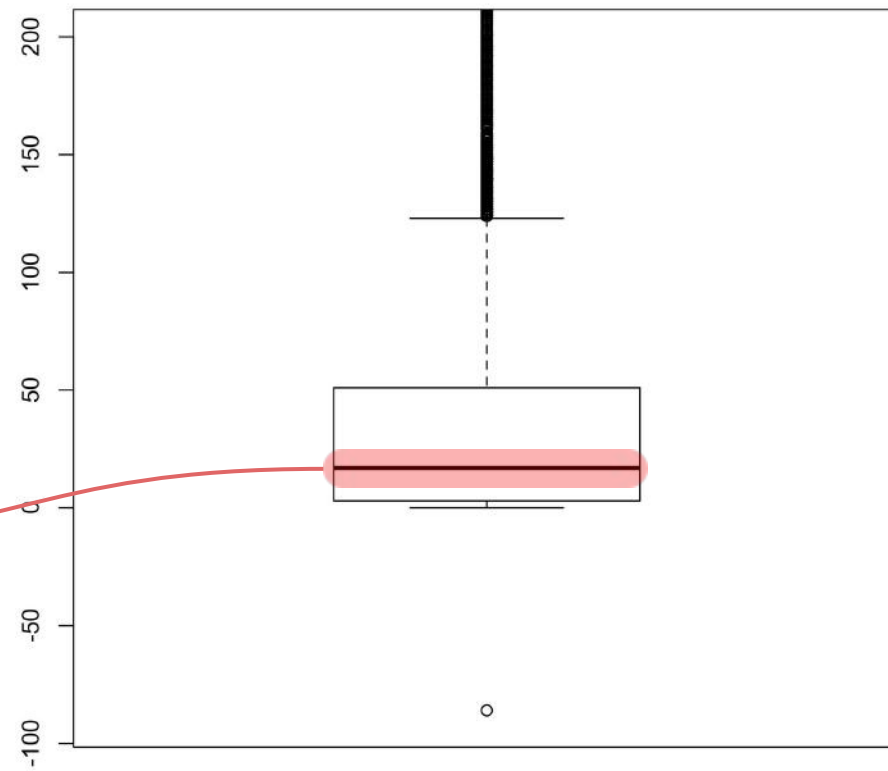


Excursion: Learning How to Read Box Plots

median

```

X0
Min.   : -86.00
1st Qu.:   3.00
Median :  17.00
Mean   :  38.85
3rd Qu.:  51.00
Max.   :1329.00
    
```

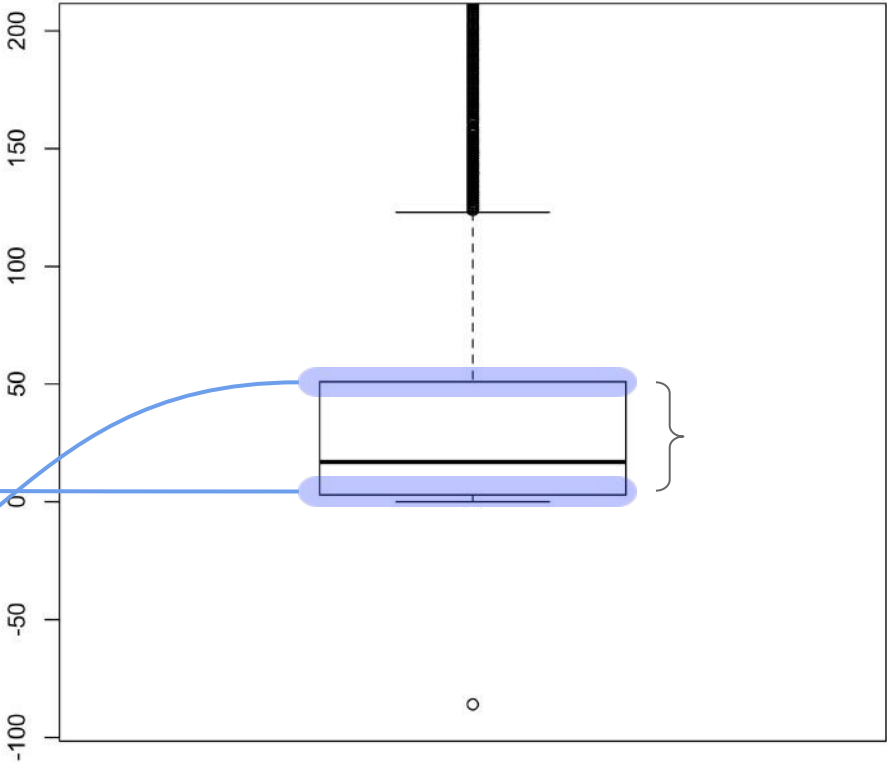


Excursion: Learning How to Read Box Plots

first quartile (Q_1)

third quartile (Q_3)

	X0
Min.	-86.00
1st Qu.:	3.00
Median	17.00
Mean	38.85
3rd Qu.:	51.00
Max.	1329.00



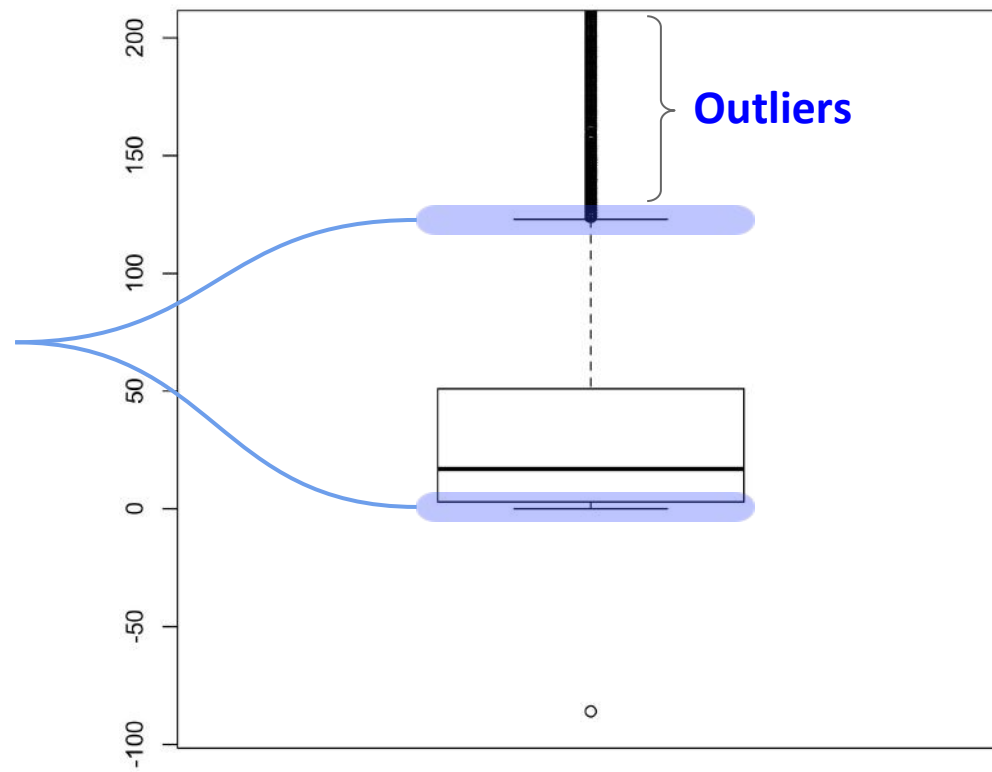
Excursion: Learning How to Read Box Plots

Whiskers

outlier

Whiskers

	X0
Min.	: -86.00
1st Qu.:	3.00
Median :	17.00
Mean :	38.85
3rd Qu.:	51.00
Max.	:1329.00

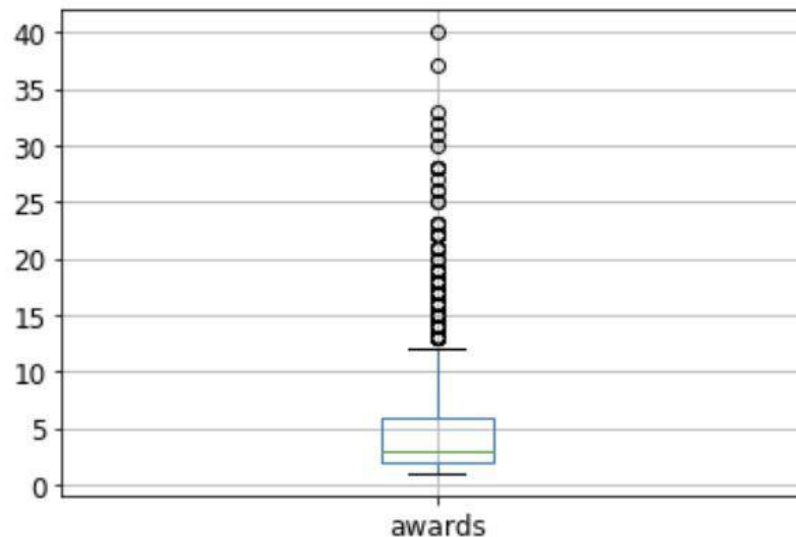


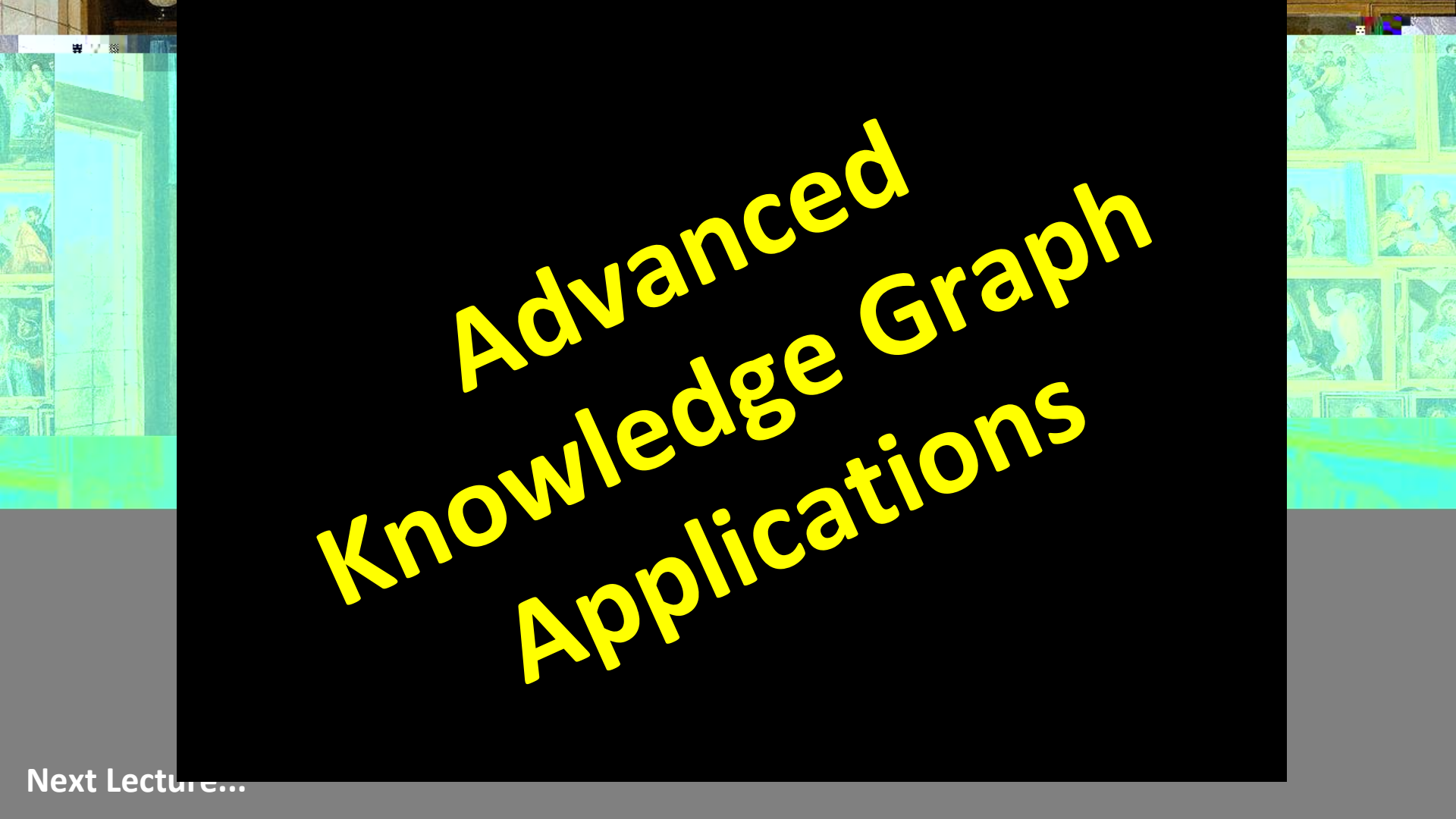
Excursion: Learning How to Read Box Plots

Analyse the Data

```
[29] physicists.boxplot('awards')
```

```
> <matplotlib.axes._subplots.AxesSubplot at 0x7f3c00578f28>
```



The background of the slide is a collage of classical art. On the left, there is a vertical strip showing a detail of a classical building facade with a window and a fresco of a seated figure. On the right, there is a grid of several classical paintings, including a group of figures in a landscape, a seated figure, and a figure in a dynamic pose. The central text is overlaid on a black rectangular area.

Advanced Knowledge Graph Applications

Next Lecture...

Picture References:

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